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Improving prediction selectivity for on-line near-infrared monitoring of components in etchant solution by spectral range optimization

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ABSTRACT

The components (H_3PO_4, HNO_3, CH_3COOH and water) in an etchant solution have been accurately measured in an on-line manner using near-infrared (NIR) spectroscopy by directly illuminating NIR radiation through a Teflon line. In particular, the spectral features according to the change of H_3PO_4 or HNO_3 concentrations were not mainly from NIR absorption themselves, but from the perturbation (or displacement) of water bands; therefore, the resulting spectral variations were quite similar to each other. Consequently partial least squares (PLS) prediction selectivity among the components should be the most critical issue for continuous on-line compositional monitoring by NIR spectroscopy. To improve selectivity of the calibration model, we have optimized the calibration models by finding selective spectral ranges with the use of moving window PLS. Using the optimized PLS models for each component, the resulting prediction accuracies were substantially improved. Furthermore, on-line prediction selectivity was evaluated by spiking individual pure components step by step and examining the resulting prediction trends. When optimized PLS models were used, each concentration was selectively and sensitively varied at each spike; meanwhile, when whole or non-optimized ranges were used for PLS, the prediction selectivity was greatly degraded. This study verifies that the selection of an optimal spectral range for PLS is the most important factor to make Teflon-based NIR measurements successful for on-line and real-time monitoring of etching solutions.

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1. Introduction

The feasibility of near-infrared (NIR) spectroscopy [1,2] for the measurements of important components of etching solutions directly through Teflon tubing (PFA: perfluoroalkoxy fluorocarbon) has recently been demonstrated by our group [3–5]. Additionally, direct Raman measurement through Teflon tube has been also demonstrated [6]. Direct NIR spectroscopic measurement without any process system modifications would be beneficial as Teflon tubing is currently being used at wet stations [7]. To consolidate the reliability of NIR measurements through Teflon tubes, we have further investigated the variation of partial least squares (PLS) [8,9] prediction performance using different Teflon tubes [4] and selectivity of PLS calibration using PCSA (pure component selectivity analysis) [5]. Although feasible results have been demonstrated in previous studies, the presented NIR spectroscopic strategy has not been expanded to measure components in an on-line

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fashion that would be practical in a real industrial setting. Moreover, on-line prediction selectivity should be critically addressed in order to provide actual applicability for optimal process control and operation since the resulting peaks of components, especially from inorganic acids, are highly overlapping without distinct features, resulting from indirect water band perturbation rather than their own absorption [10,11].

In this paper, we have presented the capability of the NIR measurement scheme (direct transmission through Teflon lines) for on-line monitoring of the components in an etchant solution. With this objective, we built a pilot-scale chemical wet station that can simulate a real cleaning bench and generate continuous flow. NIR spectra were then directly collected from on-line samples. We employed an metal aluminum etchant (MAE) solution (composed of H_3PO_4, HNO_3, CH_3COOH and water), one of the most frequently utilized etchants, for this study. PLS calibration models were built for each component and their on-line prediction selectivities were evaluated by spiking individual components into a mixture sample. When the spectral ranges were optimized using the method of moving window PLS [12,13], on-line prediction selectivity was significantly improved compared to the use of non-optimized models. Since the NIR features of the components were broadly overlapping with the underlying water band, the selection of an optimal spectral range was critical for achieving reliable prediction selectivity. Of particular importance was the spectral discrimination between H_3PO_4 and HNO_3 since both spectral variations based on concentration change in NIR range were quite similar to each other and deficient of selectivity. Overall, the concentrations of each component were measured in an on-line manner with good accuracy and selectivity by properly optimizing the spectral range for PLS.

2. Experimental

2.1. Sample preparation and pilot chemical wet bench

Ten standard samples were prepared by mixing appropriate amounts of H_3PO_4, HNO_3, CH_3COOH and water. The concentration of each component was determined by generating 10 random numbers in a given concentration range. The concentrations of H_3PO_4, HNO_3, CH_3COOH and water ranged from 69.7–49.7, 8.0–1.2, 10.0–2.0 and 43.3–16.2 wt.%, respectively. The densities of H_3PO_4, HNO_3, CH_3COOH and water were 1.69, 1.41, 1.05 and 1.00 g mL^{-1}, respectively. The concentration ranges of the standard mixtures were slightly wider than the concentration ranges in the actual process. The chemicals were purchased from a common supplier.

Fig. 1 shows the schematic diagram of the pilot chemical bench built for this study. It is designed to control temperature and flow rate, and circulate an etchant sample continuously through 1/2 or 3/4 in. Teflon tubes. The volume of the reservoir (also made of Teflon) is 5 L. The fiber optic probe, designed to transmit NIR radiation through the Teflon tubes, is installed in such a way as to hold each Teflon tube firmly in place.

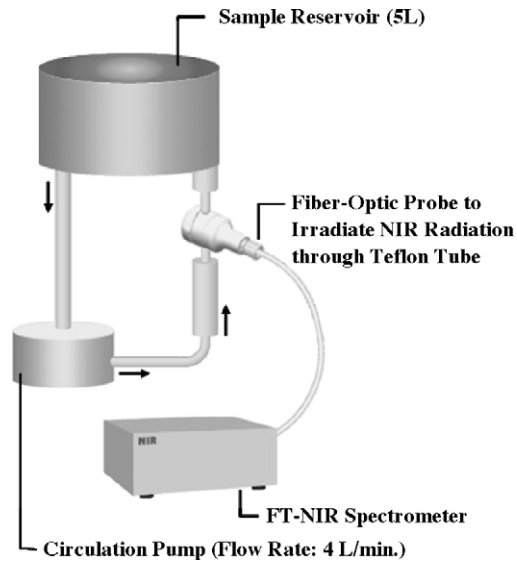

Fig. 1 – Schematic diagram of pilot chemical bench for spectral collection.

2.2. NIR spectra and data processing

NIR spectra were collected using the spectral collection device (developed by ABB Bomem Inc., Quebec City, Quebec, Canada) presented in our previous publications [3]. NIR radiation from the optical fiber is illuminated perpendicularly to the center of the Teflon tube and the transmitted radiation is focused into the receiving optical fiber. An ABB Bomem fiber optic-based FT-NIR spectrometer equipped with a tungsten-halogen source and a Germanium detector was used. NIR spectra were collected through a 3/4 in. Teflon tube while a sample was flowing. Twenty on-line NIR spectra were collected consecutively for each sample (acquisition time: approximately 1 min for each spectrum). All NIR spectra corresponded to an accumulation of 128 scans with a resolution of 32 cm^{-1} (collected at 39 °C).

A total of 200 spectra (10 samples) was divided into 140 spectra (seven samples) for the calibration set and 60 spectra (three samples) for the prediction set to validate the calibration model. All spectral calculations including PLS regression and spectral range optimization were accomplished using Matlab Version 7.0 (MathWorks Inc., MA, USA).

3. Results and discussion

3.1. NIR Spectral Features

When NIR spectra are collected through 3/4 in. Teflon tubes, the 11,000–7400 cm^{-1} range is the only choice available for the analysis of etchant solutions that usually contain diverse organic and inorganic acids in water. Fig. 2 shows 20 NIR spectra (11,000–7400 cm^{-1} range) of one selected sample (H_3PO_4:HNO_3:CH_3COOH:water = 54.30:8.00:4.08:33.62 wt.%). These spectra were collected consecutively on-line. The two major bands are located around 10,300 and 8600 cm^{-1}. The 10,300 cm^{-1} and 8600 cm^{-1} bands correspond to the second

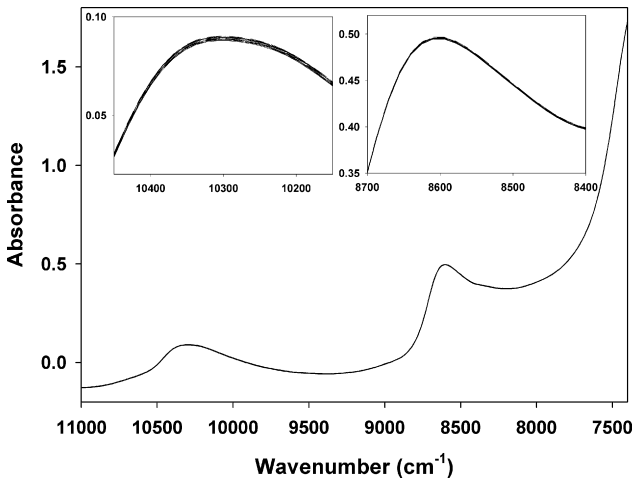

Fig. 2 – Twenty NIR spectra (11,000–7400 cm^{-1} range) of one selected sample (H_3PO_4:HNO_3:CH_3COOH:water = 54.30:8.00: 4.08:33.62 wt.%).

overtone OH band and the combination OH band, respectively. The bands are also enlarged in the same figure. As shown, the spectral feature is dominated by these two OH bands due to high molar absorptivity. Overall, the spectra are fairly repeatable by simple visual examination.

To investigate spectral variations more systematically, binary mixtures of each component and water were prepared, and corresponding NIR spectra were examined. Fig. 3 shows the NIR spectra of binary mixtures of each component in water. Only the 9000–7700 cm^{-1} range is displayed since it shows the better selectivity compared to that of the 11,000–9000 cm^{-1} range. Fig. 3(a) and (b) shows the four different concentrations of H_3PO_4 and HNO_3, respectively. As shown, the patterns of spectral variation based on concentration change for H_3PO_4 and HNO_3 are quite similar to each other. The spectral changes observed are not mainly from absorption of H_3PO_4 or HNO_3, but from the perturbation (or displacement) of water bands since these are very weak or non-NIR absorbers. That is why the magnitude of absorption is decreased in both cases, even with the increase of concentration.

The pattern of spectral variation from the CH_3COOH concentration change is generally similar to those from H_3PO_4 and HNO_3 because the spectral feature of water is still dominant due to its high concentration (refer to Fig. 3(c)). However, closer examination reveals that the spectral variation of CH_3COOH in the 8600–7700 cm^{-1} range is quite different from those of H_3PO_4 and HNO_3. This is clearly due to the difference in molecular structure of CH_3COOH (containing methyl and carbonyl groups) in spite of the fact that the previous study [10] did not show the spectral feature of CH_3COOH to be distinct in the NIR region.

To investigate the absorption features of H_3PO_4 and HNO_3 in the NIR range, gas phase spectra were collected for both analytes using high temperature gas cells (custom made, 10 cm pathlength with sapphire window and sample injection port using a syringe). Three hundred microliters of 85% H_3PO_4 and 70% HNO_3 were injected into a gas cell and the

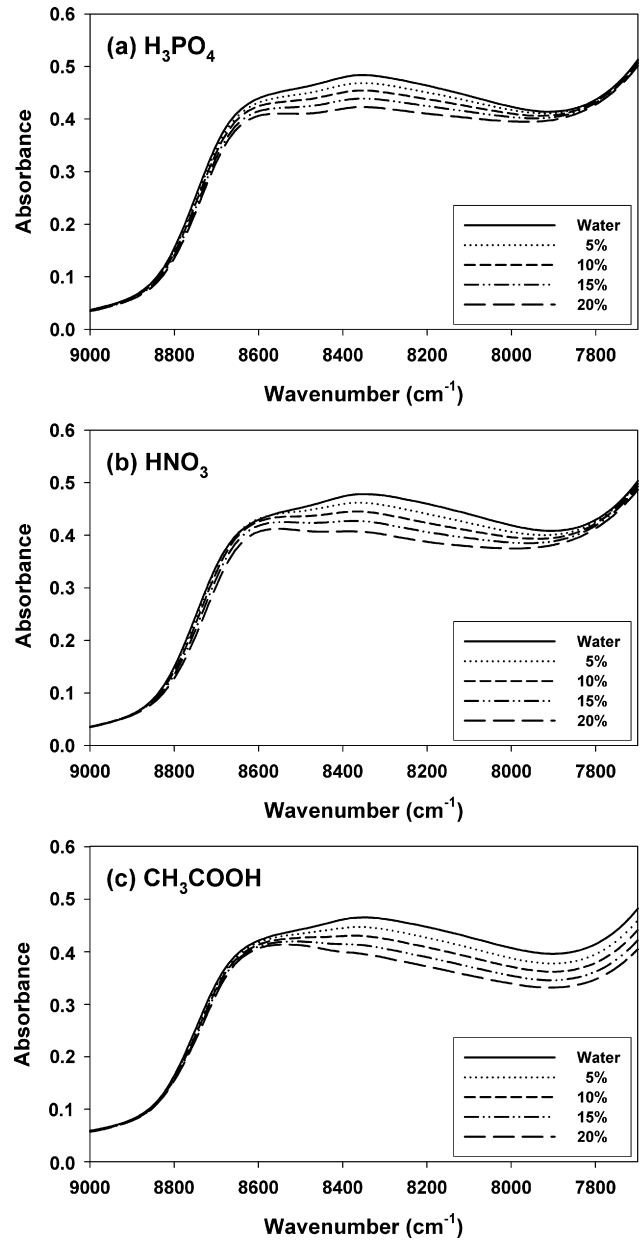

Fig. 3 – NIR spectra (9000–7700 cm^{-1} range) of water and 5, 10, 15, and 20 wt.% of H_3PO_4 (a), HNO_3 (b), and CH_3COOH (c).

corresponding NIR spectra were collected at 300 and 150 °C, respectively. Fig. 4 shows gas phase NIR spectra of 85% H_3PO_4, 70% HNO_3 and water [14,15]. The sharp rotational spectral features of water are clearly observed in the 10,700–10,300 and 9100–8400 cm^{-1} ranges. By comparison with the spectral feature of water, no noticeable spectral feature is observed for H_3PO_4; meanwhile, there are two observable bands for HNO_3. The two bands centered around 10,170 and 8200 cm^{-1} correspond to the second overtone OH band of HNO_3 and the combination band between the OH stretch and the NO_2 symmetric stretch, respectively. Although HNO_3 has a unique band at 8200 cm^{-1}, the major spectral variation could have originated from the variation of the OH band of water. This is probably due to the significant band broadening in a liquid

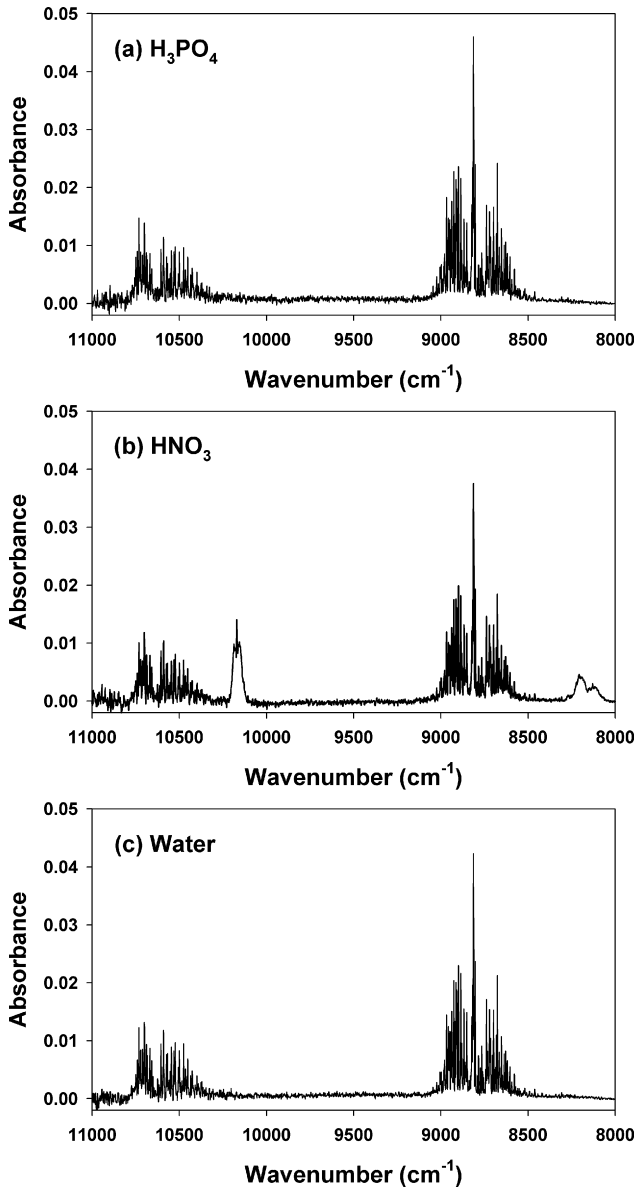

Fig. 4 – Gas phase NIR spectra of 85% H_3PO_4 (a), 70% HNO_3 (b), and water (c).

phase as well as the low molar absorptivity HNO_3 peak. Based on the examination of gas phase spectra of H_3PO_4 and HNO_3, the spectral variation from absorption by these components should be insignificant in an etchant solution. As discussed, the perturbation or displacement of water bands should be the major source of spectral variation for quantification.

3.2. *Range optimization for PLS calibration*

A partial least squares (PLS) regression was used to build calibration models for each component. Initially three different spectral ranges were utilized: the 11,000–7400, 11,000–9300 and 9300–7400 cm^{-1} ranges. The 11,000–7400 range incorporates the whole spectral range. The 11,000–9300 and 9300–7400 cm^{-1} ranges use the regions of the second overtone OH band and the combination OH band, respectively. The overall PLS calibration results are summarized in Table 1. Numbers in parentheses correspond to the number of factors used. An SEC (standard error of calibration) was estimated using the calibration set and an SEP (standard error of prediction) was calculated by predicting samples in the prediction set. The number of factors was determined by an *F*-test.

For CH_3COOH and water, which have relatively distinct features, the best prediction results were obtained using the 11,000–7400 cm^{-1} range by incorporating full spectral information, while, for H_3PO_4 and HNO_3, the use of the more selective region (9300–7400 cm^{-1}) resulted in improved prediction performance. Higher spectral selectivity should be required for these components due to low spectral sensitivity based on concentration changes, as shown in Fig. 3. The resulting SEPs for each component are larger than the corresponding SECs. It is clearly evident that the developed PLS models do not perfectly describe the variation presented in the prediction set. This should be improved for long-term accurate NIR monitoring of each component.

Since the spectral variations according to the concentration changes between H_3PO_4 and HNO_3 are fairly similar to each other, the proper selection of an optimal spectral range would be crucial in order to selectively differentiate these two components. The selectivity of CH_3COOH could also be carefully considered due to its low spectral selectivity, even with a different molecular structure. The use of the whole or wide spectral range for PLS could be advantageous since all the information can be utilized. However, this might not always be realized when the spectral features of a given ana-

Table 1 – PLS calibration results for all four components using the three different spectral ranges (11,000–7400, 11,000–9300 and 9300–7400 cm^{-1} range) (unit: wt.%)

Components	Spectral range					
	11,000–7400 cm^{-1}		11,000–9300 cm^{-1}		9300–7400 cm^{-1}	
	SEC	SEP	SEC	SEP	SEC	SEP
H_3PO_4	0.10 (8)	0.91	0.20 (8)	1.50	0.18 (7)	0.66
HNO_3	0.13 (8)	0.71	0.14 (7)	0.39	0.32 (7)	0.57
CH_3COOH	0.20 (6)	0.39	0.27 (6)	5.80	0.14 (6)	1.21
Water	0.07 (6)	0.64	0.13 (6)	0.81	0.08 (6)	0.75

The numbers in parentheses correspond to the number of factors used.

Table 2 – PLS calibration results for all four components using the optimized spectral ranges for each component (unit: wt.%)

Components	SEC (wt.%)	SEP (wt.%)	Optimal range (cm^{-1})
H_3PO_4	0.07 (7)	0.13	8600–7440
HNO_3	0.10 (7)	0.16	9580–7580
CH_3COOH	0.07 (5)	0.10	8840–8360
Water	0.11 (6)	0.05	8840–8500

The numbers in parentheses correspond to the number of factors used.

lyte broadly overlap with those from other components. There exists, possibly, an optimal spectral range that yields improved calibration results.

To find the optimal spectral ranges for each component, we have used the same moving window PLS as performed in the previous paper [12,13]. The main strategy was to perform a PLS at all possible combinations of spectral ranges and find the spectral range that yields a minimum of error. The window size (number of spectral points) was varied from 50 to 3000 cm^{-1} at increments of 20 cm^{-1}. In a given window size, the window was moved from low to high wave numbers in 20 cm^{-1} steps until the whole spectral range (11,000–7400 cm^{-1}) was covered. A total of 9497 spectral ranges of different window sizes and different spectral ranges were evaluated.

To perform a moving window PLS, each original calibration set was further divided into a new calibration set (100 spectra of five samples) and a new validation set (40 spectra of two samples). At each spectral range, a PLS was performed using the new calibration set and the resulting SEP from the new validation set was evaluated to determine the optimal spectral range (corresponding to the lowest SEP). Then, a PLS was again performed using the original calibration set within the optimal spectral range and the final SEP was calculated by predicting samples in the original validation set. The overall results are summarized in Table 2. The numbers in parentheses correspond to the number of factors used.

For all four components, the prediction accuracies were significantly improved compared to the use of the non-optimized spectral range as shown in Table 1 by finding each optimal spectral range. In the case of using optimal ranges for each component, the differences between the SEC and the SEP were greatly reduced and similar to each other. With the use of PLS models developed using the optimal spectral range, the spectral variation related to a given analyte was selectively described with minimal spectral influences from other components. The corresponding scatter plots showing correlations between reference and NIR analyses for each component are shown in Fig. 5. The open and filled circles represent calibration and prediction data, respectively. In all cases, the calibration and prediction data had a good correlation with the reference data.

3.3. *On-line prediction selectivity*

To further demonstrate the selectivity of each optimized PLS model, on-line prediction performances were tested by spiking pure components to a mixture solution and evaluating the prediction trends. Initially, 5 L of a mixture solution (H_3PO_4:HNO_3:CH_3COOH:water = 59.81:5.00:4.01:31.18 wt.%) was prepared and on-line NIR spectra were continuously collected. Then, concentrations of each component were predicted using both non-optimized PLS models (9300–7400 cm^{-1} range for H_3PO_4 and HNO_3, 11,000–7400 cm^{-1} range for CH_3COOH and water) and optimized PLS models as described in Table 2. The overall on-line prediction results using non-optimized and optimized PLS models are presented in Fig. 6.

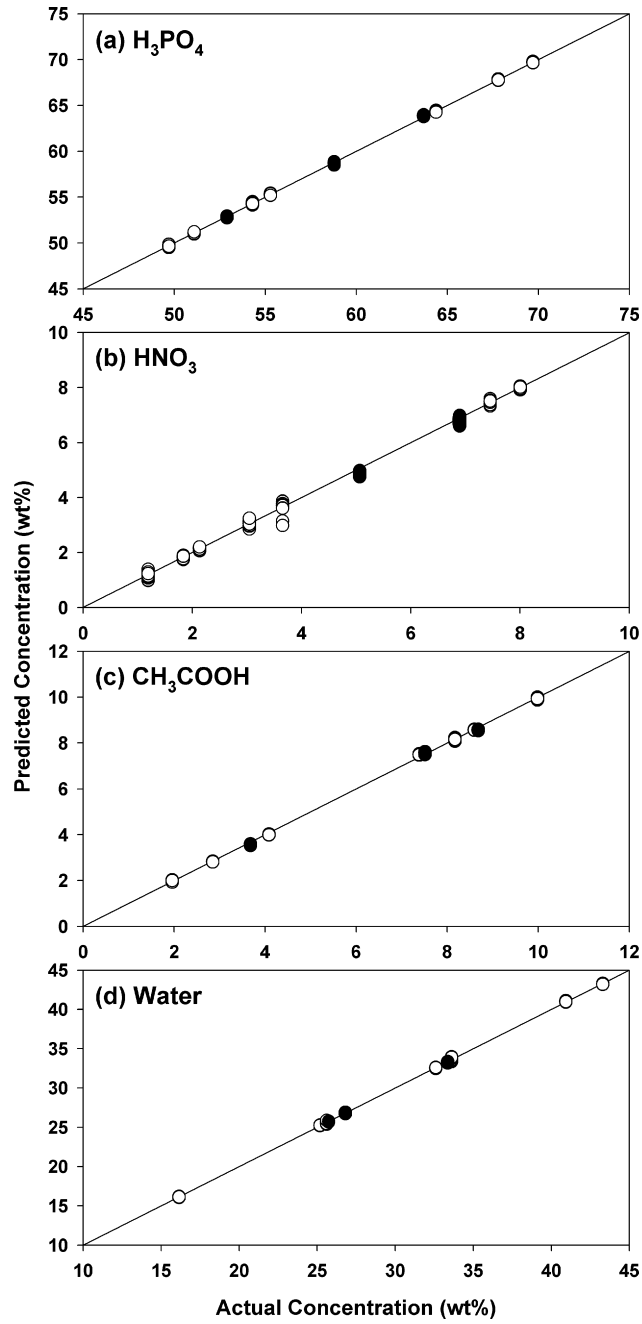

Fig. 5 – Scatter plots showing correlations between reference and NIR analyses for each component. Open and filled circles represent the calibration and prediction data, respectively.

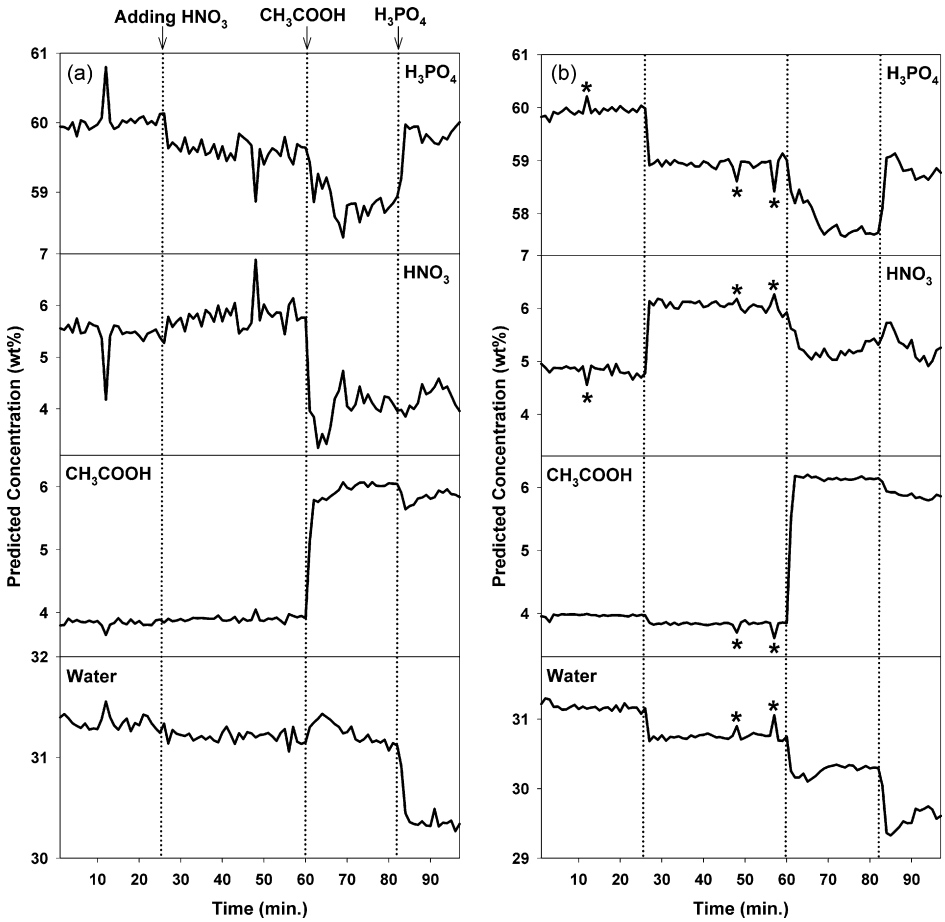

Fig. 6 – On-line prediction performances using non-optimized (a) and optimized (b) PLS models by spiking a pure component to a mixture solution. The asterisks correspond to sudden spikes in prediction trends from abnormal spectral changes caused by bubbles present in a sample stream.

Approximately 25 min after initial sample circulation, 70 mL of 70% HNO_3 was added. The sudden increase of HNO_3 concentration is clearly observed just after spiking when the optimized model of HNO_3 is used; meanwhile, there is no significant change in predicted concentrations in the case of using the non-optimized range (9300–7400 cm^{-1}). Additionally, the concentrations of the other three components decrease simultaneously using the optimized models due to the relative dilution by the addition of HNO_3 into a mixture. The concentration variations of all four components are systematically varied after the spiking of HNO_3 using optimized PLS models; meanwhile, no similar trend is observed when non-optimized models are used.

After 60 min, 95 mL of 99% CH_3COOH was spiked. After the addition of CH_3COOH, sudden increases of CH_3COOH concentration are observed for both cases. This is because the spectral feature of CH_3COOH is relatively distinct; therefore, the selectivity of prediction is relatively immune to the selection of spectral range. When the optimized models were used, the concentration variations of the other three components are more logical in that these concentrations are decreased after the spiking of CH_3COOH. However, there is no significant change of water concentration in the case of using a non-optimized model.

Finally, 120 mL of 85% H_3PO_4 was spiked after 82 min. As in the cases of HNO_3 and CH_3COOH, the selective and logical concentration variations of H_3PO_4 and the other three components are observed using each optimized PLS model. The increase of HNO_3 concentration, even after adding H_3PO_4, is probably from the incomplete mixing of H_3PO_4 due to its high density. It would require some additional time for complete mixing; therefore, the HNO_3 concentration decreases 10 min after the addition of H_3PO_4 (around 93 min).

Three instances of apparent decreases of water concentration at three different spikes are clearly observed using the optimized model; meanwhile, only one noticeable concentration drop after the spiking of H_3PO_4 is observed with the use of the non-optimized model. The sudden spikes in the prediction trend (marked as asterisks on the right plot in Fig. 6) are from abnormal spectral changes due to bubbles present in the sample stream.

The overall results demonstrate the importance of range optimization for PLS especially when spectral features are highly overlapping or indirect quantification using a shift or intensity change of the water peak is attempted. For the successful on-line monitoring of components in etchant solutions, which usually contain hydrogen peroxide (H_2O_2)

and various acids, a PLS model should be carefully developed to maintain prediction selectivity.

4. Conclusions

An on-line monitoring of components in MAE solutions by direct measurement through Teflon tubes using NIR spectroscopy has been successfully demonstrated. The selection of an optimal spectral range for PLS calibration was inevitable for improving the prediction selectivity and stability, since the corresponding spectral features of each component were highly overlapping without distinct spectral characteristics. To take advantage of NIR spectroscopy such as speed of analysis, multi-component analysis and superior repeatability for the monitoring of etchant solutions, the procedure for optimizing PLS models should be necessary rather than simple use of whole or wide spectral range that is frequently repeated in real industrial fields.

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